SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
(Common to all Branches)

Mode of Exam
OFFLINE

DEPARTMENT OF COMPUTING TECHNOLOGIES
SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu
Academic Year: 2023-24(ODD)

Test: CLA-1 **Date: 18-08-2023**
Course Code & Title: 21CSS201T& Computer Organization & Architecture **Duration: 100 Minutes**
Year & Sem: II &IV **SET C** **Max. Marks: 50**

Course Articulation Matrix:

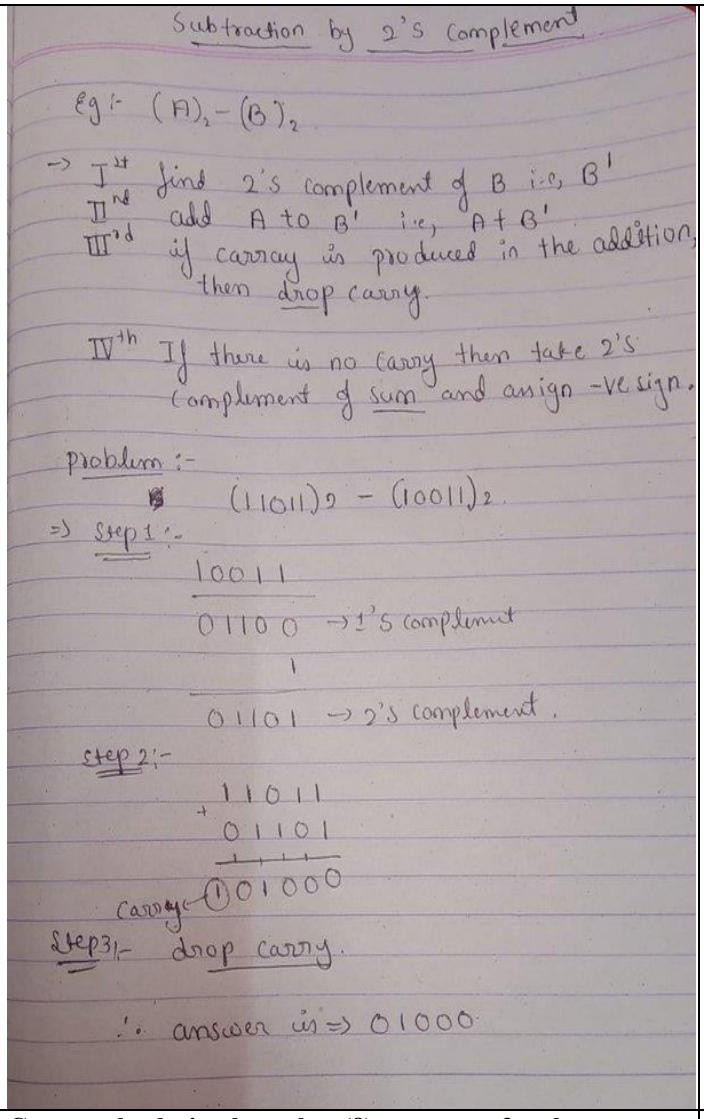
Course Learning Outcomes (CLO)	At the end of this course, learners will be able to:	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO-1	Identify the computer hardware and how software interacts with computer hardware	H	M	-		-	-	-	-	-	-	-	-
CO-2	Apply Boolean algebra as related to designing computer logic, through simple combinational and sequential logic circuits	H	H	-		-	-	-	-	-	-	-	-

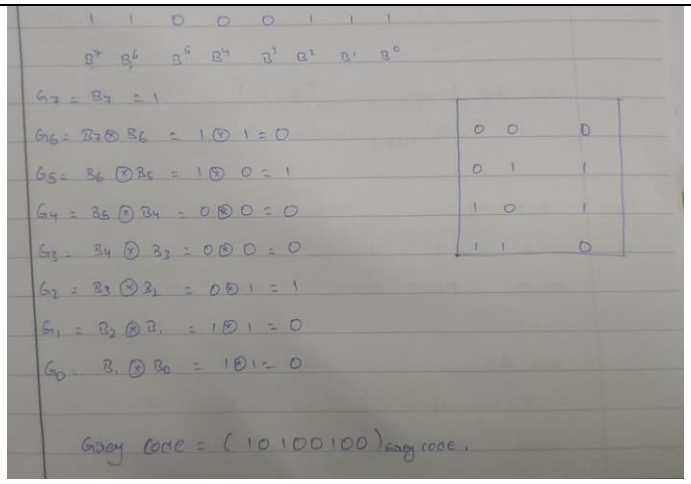
Part – A
(10 x 1 = 10 Marks)

Instructions: Answer all

Q. No	Question	Marks	BL	CO	PO	PI Code
1	The output of a NOR gate is HIGH if _____. a.all inputs are HIGH b.any input is HIGH c.any input is LOW d.all inputs are LOW	1	1	1	2	2.1.2
2	Base of octal number system is a.10 b.2 c.8 d.16	1	1	1	2	2.1.2
3	Decimal equivalent of hexadecimal no. $(44A)_{16}$ is : a.825 b.1098 c.870 d.1100	1	3	1	2	2.1.2
4	Binary equivalent of $(542)_{10}$ is : a.101010101 b.1110101101 c.101011001 d.1000011110	1	3	1	2	2.1.2
5	Convert the binary number 1011011 to hexadecimal. a.5A b.5D c.5B d.5C	1	3	1	2	2.1.2

6	**Octal representation of $(0101)_2$ is :** **a.5** **b.3** **c.4** **d.6**	**1**	**2**	**1**	**2**	**2.1.2**
7	**Cache memory acts between** **a.CPU and RAM** **b.RAM and ROM** **c.CPU and Hard Disk** **d.ROM and CPU**	**1**	**2**	**2**	**2**	**2.1.2**
8	**Which of the following is lowest in memory hierarchy** **a.Cache memory** **b.Registers** **c.RAM** **d.Secondary memory**	**1**	**1**	**2**	**2**	**2.1.2**
9	**Which of the following is the correct BCD representation of the decimal number 89?** **a.10001001** **b.10011001** **c.10100001** **d.10100101**	**1**	**1**	**2**	**2**	**2.1.2**
10	**Which instruction is used for moving the data from accumulator to memory?** **a.Move B** **b.Store D** **c.Load D** **d.PUSH A**	**1**	**1**	**2**	**2**	**2.1.2**
Part – B **(4 x 4 = 16 Marks)** **Instructions: Answer any 4**						
11	**Convert the given binary number $(01100100)_2$ to ASCII text conversion.** $(01100100)_2 = (100)_{10}$ 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0 128 64 32 16 8 4 2 1 0 1 1 0 0 1 0 0 64 + 32 + 4 =100	**4**	**3**	**1**	**2**	**2.1.2**
12	**Utilize 1's complement and 2's complement method for the following subtraction $(11011)_2 - (10011)_2$**	**4**	**3**	**1**	**2**	**2.1.2**

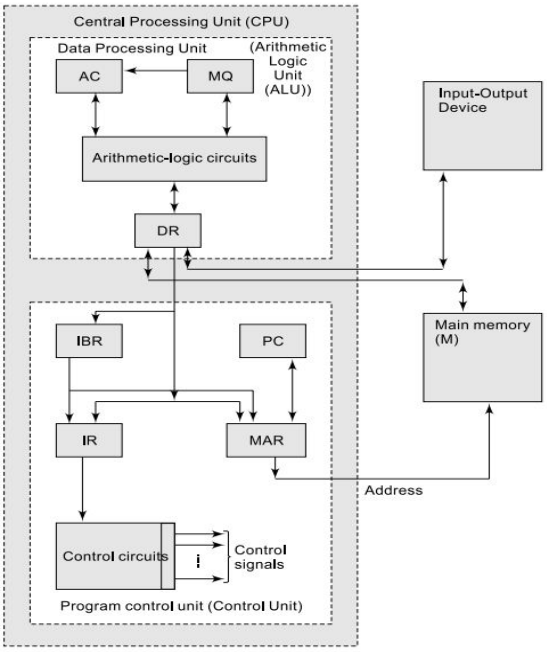
						
13	**Convert the decimal number $(8)_{10}$ to excess 3 code.** **Answer:** Binary number of 8: 1000 Binary number of 3: 0011 Adding results : 1011	**4**	**2**	**1**	**2**	**2.1.2**
14	**What are the functional units of a computer?** **Answer:** A computer in its simplest form comprises five functional units namely input unit, output unit, memory unit, arithmetic & logic unit and control unit.	**4**	**1**	**2**	**2**	**2.1.2**
15	**A computer has 128 MB of memory. Each word in this computer is eight bytes. How many bits are needed to address any single word in memory?** **Answer:** The memory address space is 128 MB, which means 227. However, each word is 8 (23) bytes, which means that you have 224words. This means you need log2224or 24 bits, to address each word.	**4**	**2**	**1**	**1**	**1.3.1**
	Part – C **(2 x 12 =24 Marks)**					
16(a)	**Obtain the following conversion** **i)$(11000111)_2$ to Gray Code.(3 Marks)**	**12**	**3**	**1**	**2**	**2.1.2**

	 **ii) Convert Excess-3 (36) to binary(3 Marks)** **Answer:** Step 1: Add '3' to the individual given decimal The Excess-3 code of given decimal (36) is $(0110\ 1001)_{Excess-3}$ **iii)Perform BCD addition for the decimal 324.13 and 846.46 (6 Marks)** **Answer:** 324.13+846.46 BCD addition (Here we have consider this numbers as decimal value) After removing 2 precision, Numbers are 32413 and 84646 For A+B 1. Add each digit of A and B using binary addition 2. If sum of two digits is more than 9 then result is Invalid BCD and add 6 to the result, Otherwise result is valid BCD. 3. If carry then add it to the next bits Add 32413 and 84646 using BCD addition BCD code for 32413 :0011 0010 0100 0001 0011 BCD code for 84646 :1000 0100 0110 0100 0110 Addition : 1011 0110 1010 0101 1001 If Invalid BCD then add 6 :0110 0110 Addition : 10001 0110 10000 0101 1001 Remaining bits except carry :10001 0110 0000 0101 1001 Carry : 1 Addition : 10001 0111 0000 0101 1001 BCD value : 11 7 0 5 9 So final answer of BCD addition is 1170.59 **(Or)**					
16(b)	**i) Compute 2's complement for the following binary values: a. 10011010 b. 00011001010 (4 Marks)** a.1's complement representation: 0110 2's complement representation: 0111 b.1's complement representation: 1110 2's complement representation: 1111 **ii) Divide 11010 by 101 (4 Marks)**	**12**	**2**	**1**	**2**	**2.1.2**

	Answer: 101) 1 1 0 1 0 (101 1 0 1 _____ 0 0 1 1 0 - 1 0 1 _____ 0 0 1 Ans: $(101)_2$ **iii) List out the steps to implement the boolean expression using NAND gate. (4 Marks)** **Answer:** a. Represent function using AND and OR gate b. Convert all OR gates to invert OR and all AND gates to NAND gates c. Check all the bubbles in the representation, if not balanced, insert an inverter on the same line d. Convert all invert OR and NOT to NAND gate					
17(a)	**i) A gaming controller sends input data to a console using Gray code. The last transmitted Gray code was 1101, indicating a button press. The current received Gray code is 1011. Determine which button was released after the press based on the given Gray codes. (4 Marks)** **Answer :** To determine which button was released after a button press, we need to compare the last transmitted Gray code with the current received Gray code and identify the bit position where a change occurred. The bit position that changed will correspond to the button that was released. Given the last transmitted Gray code: 1101 Current received Gray code: 1011 Let's compare the two Gray codes bit by bit: • 1st bit (leftmost): No change (1) • 2nd bit: Change from 1 to 0 (released button) • 3rd bit: Change from 0 to 1 (pressed button) • 4th bit (rightmost): No change (1) Based on the comparison, the 2nd bit changed from 1 to 0, indicating that the button associated with the 2nd bit was released after the press. So, the button that was released after the press is the button corresponding to the 2nd bit of the Gray code. **ii) Convert the hexadecimal number D4C.82C to its binary, decimal and octal equivalent using the iterative method. Show all the steps. (8 Marks)**	**12**	**2**	**2**	**2**	**2.1.2**

- **Step 1: Convert Hexadecimal to Binary Hexadecimal: D4C.82C**
- Converting the whole number part (D4C) to binary: D = 13 (decimal) = 1101(binary) 4 = 4 (decimal) = 0100 (binary) C = 12 (decimal) = 1100 (binary)
- So, the whole number part in binary is 110101001100.
- Converting the fractional part (82C) to binary: 8 = 8 (decimal) = 1000 (binary) 2 = 2 (decimal) = 0010 (binary) C = 12 (decimal) = 1100 (binary)
- So, the fractional part in binary is 100000101100.
- Combining the whole number and fractional parts: D4C.82C (hexadecimal) = 110101001100.100000101100 (binary).
- **Step 2: Convert Binary to Decimal Binary:** 110101001100.100000101100
- For the whole number part: 1 * 2^11 + 1 * 2^10 + 0 * 2^9 + 1 * 2^8 + 0 * 2^7 + 1 * 2^6 + 0 * 2^5 + 0 * 2^4 + 1 * 2^3 + 1 * 2^2 + 0 * 2^1 + 0 * 2^0 = 2048 + 1024 + 256 + 64 + 32 + 8 + 4 = 4456
- For the fractional part: 1 * 2^-1 + 0 * 2^-2 + 0 * 2^-3 + 0 * 2^-4 + 0 * 2^-5 + 0 *2^-6 + 1 * 2^-7 + 0 * 2^-8 + 1 * 2^-9 + 1 * 2^-10 + 0 * 2^-11 = 0.5 + 0.0078125 + 0.0009765625 = 0.5087890625
- Adding the whole number and fractional parts: 4456 + 0.5087890625 ≈4456.5087890625
- So, the binary number 110101001100.100000101100 is approximately equal to the decimal number 4456.5087890625.
- **Step 3: Convert Decimal to Octal Decimal:** 4456.5087890625
- For the whole number part: Dividing 4456 by 8 gives us a quotient of 557 and a remainder of 0. Dividing 557 by 8 gives us a quotient of 69 and a remainder of 5. Dividing 69 by 8 gives us a quotient of 8 and a remainder of 5. Dividing 8 by 8 gives us a quotient of 1 and a remainder of 0. Dividing 1 by 8 gives us a quotient of 0 and a remainder of 1.
- Reading the remainders from bottom to top, the whole number part in octal is 10550.
- For the fractional part: Converting the fractional part to octal: 0.5087890625 * 8 =4.0703125
- So, the fractional part in octal is 0.4.
- Combining the whole number and fractional parts in octal notation: 4456.5087890625 (decimal) = 10550.4 (octal).

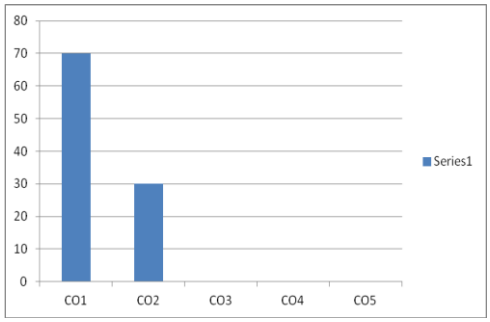
(Or)

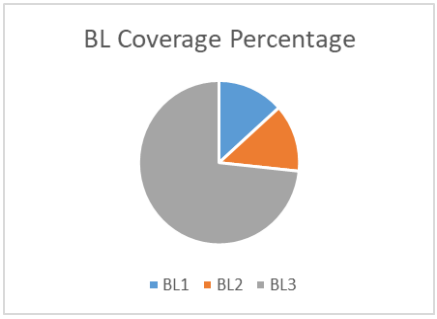
17(b)	**How can a connection be made between the processor,i/o unit and main memory to execute a simulation effectively?(12 Marks)** Diagram (4) – CPU, Main Memory Description (8) – CPU, Control unit, Input device, Output device, main memory, Registers  • CPU(Central processing unit) o CU(Control Unit) o ALU(Arithmetic and logic unit) o Registers ▪ PC	**12**	**2**	**2**	**1**	**2.1.2**

	▪ IR ▪ AC ▪ MAR ▪ MDR ▪ BUSES ● I/O Device ● MEMORY UNIT					

***Performance Indicators are available separately for Computer Science and Engineering in AICTE examination reforms policy.**

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator